CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EE) Examination

Nov-Dec 2012

CE 215 Object Oriented Programming

Date: 01.12.2012, Saturday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the followings: **[07]**

1) Which definition best describes the concept of polymorphism?
   a) Polymorphism is the technique by which an object that is used to invoke a method can actually invoke different methods, depending on the nature of the control structure.
   b) Polymorphism is the technique by which an object is used to invoke overridden methods at different times, depending on the nature of the application.
   c) Polymorphism is the technique by which an object reference is used to refer to any object created from a class that is related to the reference type by inheritance.
   d) Polymorphism is the technique by which an object's parameters are used to instantiate a particular type, depending on the nature of the parameters.
2) Usually a pure virtual function
   a) has complete function body
   b) will never be called
   c) is defined only in derived class
   d) will be called only to delete an object
3) Define Inheritance. List out all types of inheritance.
4) State any two valid differences between Structural Programming language and Object Oriented Programming language.
5) In which case is it mandatory to provide a destructor in a class?
   a) Almost in every class
   b) Class whose objects will be created dynamically
   c) Class for which two or more than two objects will be created
   d) Class for which copy constructor is defined
6) What will be the output of the following code? Assume clrscr (), getch (), return 0 and any header file are there.

```
int main()
{      int x=3,y=10,z;          z=y%x;          cout<<z;      }
```

7) To convert from a user defined class to a basic type, you would most likely use.

   a) A built-in conversion function.
   b) A one-argument constructor.
   c) A conversion function that's a member of the class.
   d) An overloaded '='operator.

Q - 2 Answer the Following Questions.

(a)	State the difference between entry controlled loop and exit controlled loop with example.	[04]
(b)	Explain Return by Reference with proper example.	[02]
(c)	What is static Data Members? Explain with example.	[04]
(d)	Explain Class to Class type conversion with one example.	[04]

OR

Q - 2 Answer the Following Questions.

(a)	What is the difference between Call by Reference and Call by value? Explain it with swapping Example.	[04]
(b)	Explain Inline Function with proper example.	[02]
(c)	What is static Member Functions? Explain with example.	[04]
(d)	What is Virtual Base Classes? Why it is needed? Explain its use with Example.	[04]

Q - 3 Answer the Following Questions.

(a)	What is the use of this pointer?	[02]
(b)	Can we pass Objects as Function Arguments? Explain it with proper Example.	[05]
(c)	Write a program that illustrates the hybrid inheritance. (Take class person and its information like name, age, DOB etc. create class student and enter its education details , create class teacher and enter its experience, create class exam and enter its marks, create class talents and create class awards. Take appropriate member variable and member functions. Implement this program. Also draw the figure of it.	[05]
(d)	What will be the output of below program?	[02]

Assume clrscr (), getch (), return 0 and any header file are there.

```
int main()
{   int var = 10;
cout<<var<<" "<<var++<<" "<<++var<<" "<<var++;   }
```

OR

Q - 3 Answer the Following Questions.

(a)	Overloading of (+) operator as a unary and binary both. We overload this operator as member function and friend function both. How many arguments it will take?	[02]
(b)	Can we return objects? Explain it with proper Example.	[05]
(c)	Write a Program to enter any number and find the reverse of a number. (without class)	[05]
(d)	What will be the output of below program?	[02]

Assume clrscr (), getch (), return 0 and any header file are there.

```
int main()
{ cout<<"Hello";     return;     cout<<"Bye";   }
```

SECTION – II

Q - 4 Answer the Following Questions.

(a) What do you mean by implicit and explicit conversion? Explain with example. [04]

(b) Define: i) scenario, ii) "is-a" relationship, iii) role name [03]

Q - 5 Answer the Following Questions.

(a) What is copy constructor? Give its use. When it is used implicitly for what purpose? [07]

(b) Write a Program to store price list of n items and to print the largest price as well as the sum of all prices also display the all items after entering details. Take necessary variables and functions by yourself. (with the use of classes) [07]

OR

Q - 5 Answer the Following Questions.

(a) What is parameterized constructor? How is it useful in dynamic initialization of objects? [04]

(b) Describe: (i) The importance of destructor. [02]

(ii) Constructor with default arguments [01]

(c) What is operator overloading? Give a programming example that overloads = = operator with its use. [07]

Q - 6 Answer the Following Questions.

(a) List and elaborate the objects and classes with their relations involved in a departmental store selling the grocery items. [07]

(b) What is aggregation? Explain it with help of proper example. Compare it with association. [07]

OR

Q - 6 Answer the Following Questions.

(a) Explain types of events and draw event trace diagram for phone call. [07]

(b) 1) Differentiate: (i) Generalization v/s Aggregation [06]

(ii) Generalization v/s Association

2) Define term state. [01]

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